Aqua-Advisor: Cost-Effective Real-Time Fish Recommendation Using Predicted Dissolved Oxygen via IoT and Machine Learning

Abstract

Specifying fish species based on the water quality conditions is a crucial measure in sustainable aquaculture. However, smallholder-level farmers usually have no way of accessing real-time environmental data. The lack of proper matching between species and their environments results in poor growth, high mortality rates, and economic losses. To overcome this, we introduce a low-cost IoT-based decision support system as an aid to select the appropriate fish species with respect to real time water quality data. This system includes a low-cost ESP32 microcontroller and temperature sensors, as well as a pH sensor that will gather real-time data, which are then used to predict Dissolved Oxygen (DO) by performing a linear regression model. Instead of using costly DO sensors, this method manages to provide fairly accurate results with a mean absolute error (MAE) of 0.42 mg/L. Afterwards, the temperature, pH, and predicted DO are fed into a rule-based engine that provides real-time fish recommendations through a Telegram bot. This solution, being designed around accessibility, aligns with UN Sustainable Development Goals (SDGs), especially SDG 2 (Zero Hunger) through efficient food production, SDG 9 (Industry, Innovation, and Infrastructure) by using IoT in the form of low-cost adoption, and SDG 14 (Life Below Water), by promoting sustainable aquaculture. Although the precision is still to be improved, this study shows that scalable data-driven solutions hold promise in leaning towards efficient small-scale farming in resource constrained environments, connecting traditional farming with smart aquaculture.

Index Terms

Aquaculture; fish recommendation systems; Dissolved Oxygen estimation; internet of things (IoT); machine learning; sustainable development; water quality monitoring

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